

BIOLOGY 414
PRINCIPLES OF BIOLOGY II
Spring, 2017

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Office Hours: Tuesday 12:00-1:00
and by appointment

Objectives: Upon completion of this course you should have acquired a basic understanding of evolution, plant organismal biology, animal organismal biology, and ecology.

Prerequisites: None

Textbooks: Campbell, et al. 2016. *Biology in Focus*, 2nd Edition. Pearson. ISBN 978-0-321-96275-1. If you used the first edition last semester, you may also use it this semester. If you are buying the book, please buy the 2nd Edition in any format you choose (hard cover, paper, or e-book).

Symbiosis. 2010. Principles of Biology Lab Manual. Pearson Learning Solutions. ISBN 978-0-536-37730-2. Can be purchased only at UNHM online bookstore.

A lab coat is required and may be purchased at the UNH-M bookstore.

Grading:	3 lecture exams @ 100 points.....	300 pts
	1 research project	100 pts
	Best 10 of 12 quizzes taken @ 10 points each.....	100 pts
	2 laboratory exams @ 30 points.....	60 pts
	8 laboratory quizzes @ 5 points.....	<u>40</u> pts
	Total Points Available.....	600 pts

All exams must be taken when scheduled. Make-up exams will be permitted only for excused absences (e.g., confirmed illness or family emergency) or if arranged in advance. Each exam only covers material since the previous exam; i.e., the second and third exams are not cumulative. Quizzes will consist of 10 multiple choice questions covering material from the previous class. They must be taken on MyCourses and must be taken by the due date. **The lowest 2 grades for quizzes that you take will be dropped** (that means if you don't take a quiz, the zero counts). There are no makeups of quizzes. Your course grade will be determined based on the percentage of points earned out of the possible 600. 90% and above will earn A- or better; 80% and above will earn B- or better; 70% and above will earn C- or better; 60% and above will earn D- or better.

Attendance:

Attendance in both lecture and laboratory is required. Excessive absences may result in a reduction of your grade.

ADA Statement:

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Disability Services Office. The Disability Services Coordinator at UNHM is Stephanie Kirylych. Stephanie can be contacted at (603) 641-4170, stephanie.kirylych@unh.edu, or in person in the Student Services Suite.

Academic Honesty:

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<https://manchester.unh.edu/current-students/academic-counseling#student-rights>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Academic Support/Tutoring:

The Center for Academic Enrichment is a resource for all students enrolled in UNH Manchester courses. Our mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. Support for other content areas is dependent on the background and competencies of current peer tutor staff. All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course. In addition, peer and professional tutors may run study groups for your class and provide drop-in math tutoring. Please visit us online at manchester.unh.edu/student/center-academic-enrichment to learn more about our services or to make an appointment with a tutor.

Human Disease Research Project:

1. Choose a human disease and submit it via the assignment tool in MyCourses.
2. Read an article in Access Science or Encyclopedia Britannica to gain general knowledge about your topic.
3. Pose a thoughtful question about your disease. Submit both the URL of the article and your question.
4. Find three journal articles using Academic Search Complete that can help you answer the question you posed. Put your references in APA format. Write a one paragraph (100 words) summary of each article.
5. Write a 3-5 page research paper where you attempt to answer your question. This paper is based on your three journal articles and your Access Science/Britannica article. Your in-text citations should be in APA format. Include your list of four references, also in APA format.
6. After seeing the tutor with a draft of your paper, make revisions and submit the final version.

All assignments must be turned in by 11:59 pm on the due date via the assignment tool in MyCourses. Late assignments will receive 0 points.

<u>Assignment</u>	<u>Due date</u>	<u>Points</u>
Submit name of disease	February 13	5
Question posed about disease; URL of Access Science article	February 20	5
Annotated bibliography in APA format	March 20	10
Meet with tutor and submit signed tutor sheet.	April 17	10
Final revised draft	April 24	70

Lecture Schedule:

<u>Date</u>	<u>Topics</u>	<u>Reading</u>
Jan 24	CANCELLED	
Jan 31	Evolution & Natural Selection	Chapters 1, 19
Feb 7	CANCELLED	
Feb 14	Population Genetics & Speciation	Chapters 21, 22
Feb 21	History, Classification, & Diversity of Life	Chapters 20, 23, 24-27
Feb 28	Plants I: Structure, Transport & Evolution	Chapters 28-30
Mar 7	EXAM I	
Mar 14	SPRING BREAK	
Mar 21	Animal Systems I: Tissues, Homeostasis, Digestion & Nutrition	Chapters 32, 33
Mar 28	Animal Systems II: Circulation & Gas Exchange	Chapter 34
Apr 4	Animal Systems III: The Immune System	Chapter 35
Apr 11	Animal Systems IV: Salt and Water Balance; Nitrogen Excretion	Chapter 32
Apr 18	EXAM II	
Apr 25	Animal Systems V: Reproduction & Development	Chapters 36
May 2	Animal Systems VI: Nervous & Sensory	Chapters 37, 38
May 9	Ecology: Ecosystems, Communities, & Populations	Chapters 40-43
May 16	EXAM III	

Laboratory Schedule:

<u>Date</u>	<u>Topics</u>	<u>Reading</u>
Feb 1, 2, 3	Introduction/Overview/Lab Safety/ Effectiveness of Hand Washing 1	page 289
Feb 8, 9, 10	Effectiveness of Hand Washing Part 2	page 289
Feb 15, 16, 17	Domain Bacteria + Domain Archaea	page 1
Feb 22, 23, 24	Kingdom Protista and Fungi	page 35
Mar 1, 2, 3	Plant Anatomy	page 73
Mar 8, 9, 10	Plant Diversity I + II	page 107+133
Mar 15, 16, 17	SPRING BREAK	
Mar 22, 23, 24	LAB EXAM I	
Mar 29, 30, 31	Animal Diversity I	page 161
Apr 5, 6, 7	Animal Diversity II	page 183
Apr 12, 13, 14	Vertebrate Anatomy I	page 217
Apr 19, 20, 21	Vertebrate Anatomy II	page 243
Apr 26, 27, 28	Vertebrate anatomy III	page 265
May 3, 4, 5	LAB EXAM II	